Computational Complexity

Reading guide and Homework for **Weeks 12&13**

1 Reading Guide

These two weeks we learned about the hardness versus randomness result of Nisan, Wigderson and Impagliazzo:

- We learned about average-case hardness, and how an average-case hard function can be used to construct a secure PRG. Sections 20.1 and 20.2.
- We mentioned locally-decodable list-decodable codes, this appears in Section 19.

2 Homework

Skill: *Be thorough.*

1. Show that, if we choose m sets $T_1, \ldots, T_m$ uniformly at random from subsets of $[k]$, then
$$\Pr[\exists i \neq j \; T_i = T_j] \leq \frac{m^2}{2^k}$$
i.e., upper-bound the probability that there exist two equal sets in the family.

2. Show that the following construction gives a pairwise independent family, as needed in the proof of the Goldreich–Levin theorem (one-way permutations implies PRGs).
   - Let $T_1, \ldots, T_m \subseteq [k]$ be some fixed family of sets, such that $T_i \neq T_j$ for all $i \neq j$.
   - Choose $s_1, \ldots, s_k \in \{0,1\}^n$ independently at random, and let
   $$r_i = \bigoplus_{j \in T_i} s_j \in \{0,1\}^n.$$
   Show that $(r_1, \ldots, r_m)$ is a pairwise independent family, i.e., for each $i, j \in [m]$, the distribution of (r_i, r_j) is uniform over $\{0,1\}^{2n}$.

3. Notice that we can think of a subset of $[k]$ as an element of $\mathbb{F}_2^k$. Then two sets T_i and T_j are different if and only if T_i and T_j are independent as vectors in $\mathbb{F}_2^n$. Now let's generalize the above to ℓ-wise independence.

   Show that if $T_1, \ldots, T_m \in \mathbb{F}_2^k = 2^{[k]}$ has the property that every ℓ sets $T_{i_1}, \ldots, T_{i_\ell}$ are independent as vectors in the linear space $\mathbb{F}_2^k$, meaning $\dim \mathsf{span}\{T_{i_1}, \ldots, T_{i_\ell}\} = \ell$, then the resulting distribution above is ℓ-wise independent, meaning, any tuple $(r_{i_1}, \ldots, r_{i_\ell})$ is uniform over $\{0,1\}^{\ell n}$.

   Finally, show that a random choice of T_i has the desired property with high probability, provided m is not much larger than k:
   $$\Pr[\exists \{i_1, \ldots, i_\ell\} \in \binom{[k]}{\ell} \dim \mathsf{span}\{T_{i_1}, \ldots, T_{i_\ell}\} < \ell] \leq \frac{\binom{m}{\ell}}{2^{k-\ell}}.$$
   (Hint: every subspace of $\mathbb{F}_2^k$ of dimension ℓ has size 2^ℓ. How likely is it that we choose a new vector and it already appears in the span of the previously chosen vectors?)

Skill: *Reason about derandomization.*

4. Do exercises 20.1–20.5 and 20.9 of the Book.